

AWS Basics

This document introduces fundamental AWS concepts and services suitable for testing document ingestion and retrieval in a RAG application.

Introduction to Cloud Computing

Cloud computing delivers computing resources over the internet on demand and enables scalability, flexibility, and cost optimization.

What is AWS?

Amazon Web Services (AWS) is a leading cloud platform that provides services for compute, storage, networking, databases, security, and artificial intelligence.

AWS Global Infrastructure

AWS Regions contain multiple Availability Zones that provide fault tolerance and high availability.

Amazon EC2

EC2 provides virtual machines that can be configured based on application requirements.

Amazon S3

S3 is highly durable object storage used for backups, media files, and static website hosting.

Amazon RDS

RDS simplifies relational database deployment and management through automated operations.

AWS Lambda

Lambda enables serverless application development without provisioning servers.

Amazon VPC

VPC provides isolated virtual networks for securely deploying AWS resources.

IAM

Identity and Access Management controls authentication and authorization across AWS resources.

Security Groups

Security Groups act as virtual firewalls that regulate network traffic.

Elastic Load Balancer

ELB distributes traffic across multiple targets to improve reliability.

Auto Scaling

Auto Scaling dynamically adjusts resources according to demand.

Amazon CloudWatch

CloudWatch offers monitoring, logging, and alerting capabilities.

AWS Pricing Models

AWS supports On-Demand, Reserved, Savings Plans, and Spot pricing models.

Shared Responsibility Model

AWS secures the cloud infrastructure while customers secure their applications and data.

Basic AWS Architecture

A common architecture combines EC2, RDS, S3, ELB, Auto Scaling, and CloudWatch inside a VPC.

Advantages of AWS

AWS provides scalability, reliability, security, and global availability.

Conclusion

AWS offers a comprehensive cloud ecosystem suitable for organizations ranging from startups to enterprises.

Sample Questions for RAG Testing

- What is Amazon EC2 used for?
- Explain the Shared Responsibility Model.
- What are AWS pricing models?
- What is the purpose of Amazon VPC?
- Which service provides object storage in AWS?